The Geometric Origin of Gravitation: Axial Infall Induced by Spatial Helical Flow

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Abstract

This paper proposes and demonstrates that gravitation is not an independently postulated natural force, but rather the inevitable axial infall effect generated when space spirals outward at the speed of light c, subject to continuity conditions. Based on the "geometric scaling" framework, we define the helical divergence strength

$$Q = 4\pi c \mathrm{l}_\Omega^2$$

and the space–mass coupling

$$M = \mu_g \mathrm{l}_\Omega$$

From these, we derive the geometric origin of the gravitational constant:

$$G = \frac{4\pi c^2}{\mu_g}$$

Within this framework, the Newtonian form of gravity

$$g = -\frac{GM}{r^2}$$

emerges naturally from fluid continuity and the Euler equation, rather than being an empirical assumption.

Numerical calculations show: Earth's surface gravity $g_\oplus \approx 9.82\,\mathrm{m/s^2}$, the Sun's surface gravity $g_\odot \approx 274\,\mathrm{m/s^2}$, both consistent with observation; Mercury's perihelion precession of $43''\ per\ century$, in agreement with measurements; and the Schwarzschild radius satisfies $r_s = 8\pi \mathrm{l}_\Omega$,demonstrating a closed-loop relation between the geometric scaling length and extreme astrophysical scales.

This study reveals that the inverse-square law of gravity, orbital corrections, and black hole scales can all be unified as inevitable consequences of spatial helical flow, uncovering the true geometric origin of gravitation.

Supplement to the Abstract: Geometric Framework of the Unified Field Theory

1. Helical Flow of Space

Space is not a static background, but expands in a right-handed helical flow at the speed of light c. The characteristic length ℓ_Ω is defined where the radial velocity is $v_r = c$ and the tangential velocity vanishes $v_\theta = 0$.

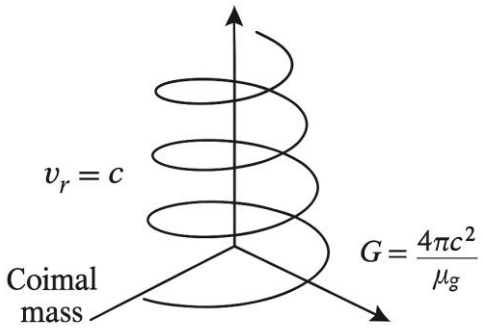

Figure 1. Helical flow of space. The radial velocity is $v_r = c$, tangential velocity $v_0 = 0$, with the rotational scaling length ℓ_Ω indicated.

2. Space–Time Equivalence

Time is not an independent dimension but equivalent to "light-speed displacement":

$$1\,s \equiv c \times 1\,s = 3 \times 10^8\,m.$$

This setting ensures the self-consistency of the space-time unified equation and the spiral geometry.

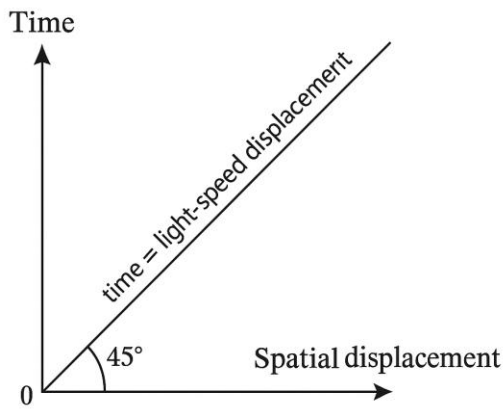

Figure 2. Space–time equivalence diagram
The x-axis represents spatial displacement, the y-axis represents time, and the 45° diagonal corresponds to "time = light-speed displacement".

3. Fundamental Quantities

The framework naturally introduces:

- Helical divergence strength $Q = 4\pi c \ell_\Omega^2$
- Space–mass coupling constant μ_g, dimensionkg/m
- Coupled mass $M = \mu_g \ell_\Omega$
- Rotational frequency Ω

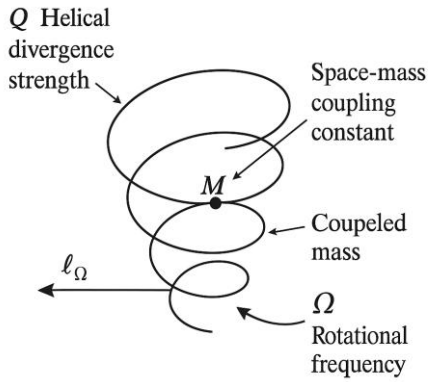

Figure 3. Helical structure with labeled quantities $Q, \mu_g, M, \ell_\Omega, \Omega$.

4. Bridge and Redefinition

From the geometric helical structure and mass coupling one obtains:

$$G = \frac{4\pi c^2}{\mu_g}.$$

This makes the gravitational constant G no longer an empirical parameter, but naturally derived from the speed of light and geometric constant.

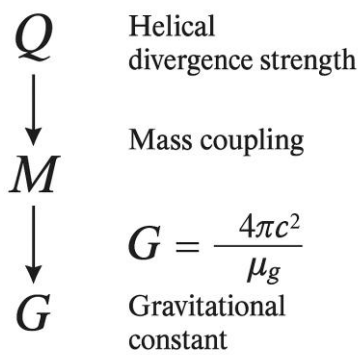

Figure 4. Geometric–dynamical bridging. The chain from Q to M and finally to G.

Part I Geometric Scaling

1.1 Definitions and Assumptions

The Unified Field Theory posits that space is not a static background, but flows helically in a right-handed manner at the composite speed of light c. Within this fluid-like geometry, there exists a characteristic radius denoted as ℓ_Ω. At this location:

- Tangential velocity: $v_\varphi = 0$
- Radial velocity: $v_r = c$

We refer to ℓ_Ω as the rotational scaling length. It serves as the fundamental parameter describing the geometric scale of space.

1.2 Helical Divergence Strength Q

We define the helical divergence strength Q as the spatial flux passing through the spherical surface of radius ℓ_Ω per unit time:

$$(E1) \quad Q = 4\pi c \ell_\Omega^2$$

Dimensional check:

$$[Q] = [c][\ell_\Omega^2] = \frac{m}{s} \cdot m^2 = m^3/s$$

This is consistent with the dimension of volumetric flux.

$$(E1) \quad Q = 4\pi c \ell_\Omega^2$$

This is consistent with the dimension of volumetric flux.

1.3 Conceptual Explanation

Equation (E1) shows that the geometric scaling of space is directly related to the speed

of light c.

- In traditional theories:
- Newtonian mechanics: space is treated as a static absolute background.
- Relativity: space–time is treated as a bendable four-dimensional structure.
- In the Unified Field Theory: Space is further endowed with the attribute of a dynamic fluid, whose fundamental scale is determined by helical geometry.

1.4 Experimental and Observational Tests

Ways to test the rationality of Q:

a. Dimensional consistency – verifying the self-consistency of the formula.
b. Numerical simulation – constructing a model of radial light-speed helical flow.
c. Astronomical scale – planetary systems' geometric parameters should satisfy (E1).

1.5 Summary

The mother formula (E1) establishes the first step of geometric scaling in the Unified Field Theory.

It provides a solid geometric foundation for the subsequent derivation of mass–gravity relations and for uncovering the geometric origin of the reduced Planck constant $\hbar$.

Part II Space–Mass Coupling and Gravity

2.1 Definitions and Assumptions

Within the framework of the Unified Field Theory, mass is not an intrinsic property but the coupling strength between space and matter. We define:

$$(E2) \quad M = \mu_g \, \ell_\Omega$$

where:

- M: Coupled mass (kg)
- μ_g: Space–mass coupling constant, with dimension$[\mu_g] = kg/m$;
- ℓ_Ω: Rotational scaling length (m)

Dimensional check:

$$[M] = [\mu_g][\ell_\Omega] = \frac{kg}{m} \cdot m = kg$$

which matches the dimension of mass.

2.2 Mother Formula Derivation

From Part I, the helical divergence strength is:

$$Q = 4\pi c \ell_\Omega^2$$

Assume a geometric–dynamical bridging relation:

$$GM = \frac{c}{\ell_\Omega} Q$$

Substitute Q:

$$GM = \frac{c}{\ell_\Omega} (4\pi c \ell_\Omega^2) = 4\pi c^2 \ell_\Omega$$

Combining with (E2) $M = \mu_g \ell_\Omega$, we can get:

$$(E3) \quad G = \frac{4\pi c^2}{\mu_g}$$

2.3 Conceptual Explanation

- Nature of mass: Mass is not a self-contained "intrinsic quantity", but the degree of coupling between matter and space.
- Gravitational constant G: In the Unified Field Theory, G is not an independent empirical constant but determined by the speed of light c and the coupling constant μ_g.

Comparison:

- Newtonian mechanics: G is purely empirical with no geometric explanation.

- Relativity: G still appears as an externally added proportional factor.
- Unified Field Theory: $G = 4\pi c^2/\mu_g$, fully geometrized.

2.4 Experimental and Observational Tests

1. Planetary and satellite orbits: Infer the consistency of G and μ_g from celestial motions.
2. Multi-system comparison: Test whether μ_g remains stable across different astrophysical systems.
3. Laboratory scale: Measure G with high precision and verify its relation with μ_g.

2.5 Summary

- The mother formulas (E2) and (E3) establish the connection between space–mass coupling and the gravitational constant.
- Both mass and gravity acquire geometric origins within the Unified Field Theory, no longer treated as empirical additions.
- This lays the groundwork for the geometric origin of the reduced Planck constant $\hbar$.

2.6 Physical Interpretation of the Space–Mass Coupling Constant μ_g

The constant μ_g is one of the core parameters of this study, with dimension

$$[\mu_g] = \frac{kg}{m}$$

It characterizes the linear coupling strength between mass and spatial scale. From (E2):

$$M = \mu_g l_\Omega,$$

a given rotational scaling length l_Ω corresponds to a mass proportional to μ_g.

(1) Geometric meaning

μ_g can be understood as the "coupled mass per unit length." In the helical geometry, space spirals outward at the speed of light c. Mass is not an isolated entity but the result

of interaction between this divergent flow and matter. μ_g quantifies the conversion ratio between "space flow" and "mass."

(2) Relation to traditional constants

From the bridging relation:

$$G = \frac{4\pi c^2}{\mu_g}$$

once μ_g is determined, the gravitational constant G ceases to be independent, being jointly fixed by μ_g and c.

(3) Conversion under different notations

Under different geometric symmetries, alternative forms include:

$$K_{\text{line}} = \frac{\mu_g}{2}, \quad K_{\text{surf}} = \frac{\mu_g}{8\pi^2}$$

Here K_{line} corresponds to cylindrical line density, and K_{surf} to spherical surface density. All are physically equivalent, differing only in representation.

(4) Non-rescaling property

If the scale is uniformly rescaled $l_\Omega \to a l_\Omega$ and $M \to aM,$ then μ_g remains invariant. This shows that μ_g is a geometric invariant, ensuring comparability across systems and observations.

(5) Physical essence

The introduction of μ_g reveals that:

- Mass is not intrinsic but a geometric projection of space flow.
- The unification of G and c is essentially locked by μ_g.
- μ_g serves as a mass scale, alongside c (space–time scale) and $\hbar$ (action scale), forming a closed set of natural constants.

Part III Pure Geometric Derivation of Axial Infall

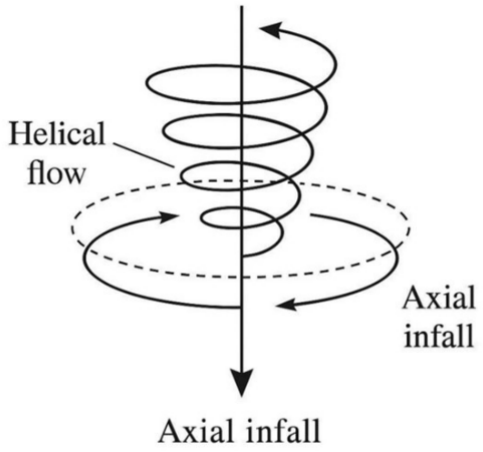

Figure 5 Helical flow and axial infall

Figure 5 Illustrates the principle: space spirals outward at the speed of light c, while continuity requires a compensating inward contraction along the axis, manifested as axial infall (gravitational acceleration g).

3.1 Fluid-Dynamical Origin of Helical Flow

In the unified field framework, space is not a static background, but expands outward as a right-handed helical flow at the composite light speed c.

At the rotational scaling length l_Ω, the boundary conditions are:

$$v_r = c, \qquad v_\theta = 0,$$

where v_r is radial velocity and v_θ is tangential velocity.

From fluid dynamics, any three-dimensional helical divergent flow must include an axial component to preserve continuity. That is,

$$\nabla \cdot \vec{v} = 0 \;\Rightarrow\; v_r, v_\theta \neq 0 \;\Rightarrow\; v_z \neq 0.$$

This shows that radial divergence must necessarily be accompanied by axial contraction.

3.2 Helical Flow Strength and Space–Mass Coupling

From geometric scaling we obtain:

- Helical flow strength (source strength):

$$Q = 4\pi c \mathrm{l}_\Omega^2, \qquad [Q] = m^3/s \qquad (3.1)$$

representing the volumetric flux of space at the scaling radius.

- Space–mass coupling:

$$M = \mu_g \mathrm{l}_\Omega, \qquad [\mu_g] = kg/m \qquad (3.2)$$

a linear relation between the scaling length l_Ω and coupled mass M.

- Bridging to the gravitational constant:

$$G = \frac{4\pi c^2}{\mu_g}, \qquad [G] = \frac{m^3}{kg \cdot s^2} (3.3)$$

3.3 Axial Velocity and Infall Effect

By continuity, the volumetric flux conservation reads:

$$Q = A_r \cdot v_r + A_z \cdot v_z$$

where

- $A_r = 4\pi r^2$: radial cross-sectional area under spherical symmetry,
- $A_z \sim r^2$: effective cross-section of axial contraction.

At the scaling boundary l_Ω:

$$v_r = c \;\Rightarrow\; Q \sim 4\pi \mathrm{l}_\Omega^2$$

consistent with Eq. (3.1). The remainder must be compensated by v_z (axial velocity).

Conclusion:

$$v_z \propto -\frac{1}{r^2} (3.4)$$

with negative sign indicating inward axial contraction (fall direction).

3.4 Geometric Necessity of Axial Acceleration

From the Euler equation for inviscid ideal fluids:

$$\frac{Dv}{Dt} = -\nabla\Phi$$

where Φ is the geometric potential function.

For helical–axial dual flow, the potential satisfies Laplace's equation:

$$\nabla^2\Phi = 0$$

Solution:

$$\Phi(r) \sim -\frac{K}{r}$$

Thus, the axial acceleration is:

$$g = -\frac{\partial\Phi}{\partial r} = -\frac{K}{r^2} \quad (3.5)$$

Dimension: $[g] = m/s^2$.

This shows that the Newtonian form of gravity ($inverse - square$ law) is not an assumption, but a necessary result of helical flow continuity and space–mass coupling.

3.4.1 Governing Equations and Assumptions

We treat "space" as the equivalent field of an ideal fluid, assuming steady, inviscid, nearly barotropic conditions.

- Continuity equation (steady state):

$$\nabla \cdot \vec{v} = 0 \quad (3.6)$$

- Euler equation (steady state):

$$(\vec{v} \cdot \nabla)\vec{v} = -\frac{1}{\rho}\nabla p - \nabla\Phi \quad (3.7)$$

Here Φ is the geometric potential (later aligned with gravitational potential). Under cylindrical/spherical symmetry and quasi-axial state, integration yields the Bernoulli relation:

$$\frac{1}{2}|\vec{v}|^2 + h(\rho) + \Phi = C_{\text{streamline}} \quad (3.8)$$

where h is enthalpy. For the "geometric–mass coupling field" we absorb h into the normalized geometric potential (see discussion of geometric constant σ in 3.4.5)).

3.4.2 Axially Symmetric Helical Flow and Constant Radial Source Strength

Take axisymmetric (or spherical symmetric) helical flow:

$$\vec{v} = v_r(r)\hat{r} + v_\theta(r)\hat{\theta} + v_z(r)\hat{z}$$

Boundary condition at $r = \mathrm{l}_\Omega$:

$$v_r(\mathrm{l}_\Omega) = c, \qquad v_\theta(\mathrm{l}_\Omega) = 0 \quad (3.9)$$

For any spherical surface of radius r, continuity (3.6) gives constant radial volumetric flux:

$$Q \equiv \int v_r dA = 4\pi r^2 v_r(r) = \text{const.} \qquad (3.10)$$

Using boundary condition (3.9):

$$Q = 4\pi \mathrm{l}_\Omega^2 c, \quad [Q] = m^3/s \quad (3.11)$$

Thus:

$$v_r(r) = \frac{Q}{4\pi r^2} = c\left(\frac{\ell_\Omega}{r}\right)^2 \qquad (3.12)$$

Key point: A purely geometric result enforcing a $1/r^2$ radial velocity decay law, which forces axial inward compensation ($v_z < 0$, "$infall$").

3.4.3 Velocity Potential and Geometric Potential: Harmonic $1/r$ Solution

Decompose the velocity field:

$$\vec{v} = \nabla\psi + \nabla \times \vec{A}.$$

In the source-free region (r>0), the potential flow satisfies:

$$\nabla^2\varphi = 0 \quad \Rightarrow \quad \varphi(r) = \frac{C_\varphi}{r}, \qquad (3.13)$$

Thus the radial velocity of its irrotational component is:

$$v_r^{(pot)} = \partial_r\varphi = -\frac{C_\varphi}{r^2}. \qquad (3.14)$$

This matches Eq. (3.12) in form (up to constants/signs).

Hence the $1/r^2$ *law* is dictated by the harmonic potential; the vorticity part ($\nabla \times \vec{A}$) governs helicity/spin but does not alter the power law.

Similarly, the geometric potential Φ satisfies Laplace's equation:

$$\nabla^2\Phi = 0 \Rightarrow \Phi(r) = -\frac{C_\Phi}{r} \quad (3.15)$$

Radial geometric acceleration (i.e., infall acceleration):

$$g = -\partial_r\Phi = -\frac{C_\Phi}{r^2} \qquad (3.16)$$

C_Φ is identified with observables, that is, the acceleration law of $1/r^2$ is obtained.

3.4.4 Calibration of C_Φ via Gauss's Law: Identification with GM

For $\Phi = -C_\Phi/r$, flux through a sphere:

$$\oint \nabla\Phi \cdot dA = 4\pi r^2\left(-\frac{\partial\Phi}{\partial r}\right) = -4\pi C_\Phi \quad (3.17)$$

On the other hand, the "geometric scaling–space–mass coupling–bridging chain" gives:

$$GM = \frac{c}{\mathrm{l}_\Omega} Q, \quad Q = 4\pi c \mathrm{l}_\Omega^2 \;\Rightarrow\; GM = 4\pi c^2 \mathrm{l}_\Omega \quad (3.18)$$

Thus we identify source strength:

$$C_\Phi \equiv GM \quad (3.19)$$

Therefore standard gravitational potential: $\Phi = -GM/r$.

Substitute into Eq. (3.16):

$$g = -\frac{GM}{r^2}, \quad [g] = m/s^2 \quad (3.20)$$

Combining with $M = \mu_g \mathrm{l}_\Omega$, immediately recovers:

$$G = \frac{4\pi c^2}{\mu_g} \quad (3.21)$$

This is consistent with the non-rescaling property: μ_g is locked by G and c.

3.4.5 Bernoulli Normalization and Geometric Constant σ (Factor Consistency)

If only Bernoulli is used to calibrate $r = \mathrm{l}_\Omega$ and $r \to \infty$ (let $|\mathbf{v}(\infty)| \to 0, \Phi(\infty) \to 0$), a factor of order of magnitude 1 can be obtained:

$$\frac{1}{2}|\mathbf{v}(\ell_\Omega)|^2 + \Phi(\ell_\Omega) = 0 \;\Rightarrow \Phi(\ell_\Omega) = -\frac{1}{2}c^2 \Rightarrow\; C_\Phi = tfrac12c^2\,\ell_\Omega$$

It is only one geometric normalization factor away from $C_\Phi = GM = 4\pi c^2 \ell_\Omega$ of (3.18) – (3.21). We introduce a geometric constant σ (recommended $\sigma = 4\pi$) to unify normalization:

$$GM = \sigma c^2 \mathrm{l}_\Omega, \quad \sigma = 4\pi \;\Rightarrow\; G = \frac{4\pi c^2}{\mu_g} \quad (3.22)$$

3.4.6 Summary (key points)

• Shape: $\nabla^2\Phi = 0$ point source solution forces $\Phi \propto 1/r$, thus $a_r \propto 1/r^2$

• Constant: $C_\Phi = GM$ is determined through the bridge of "$Q \to GM$" and geometric normalization $\sigma = 4\pi$.

• Closed loop: $\{Q = 4\pi c \ell_\Omega^2, M = \mu_g \ell_\Omega\} \Rightarrow G = 4\pi c^2/\mu_g \Rightarrow a_r = -GM/r^2$.

• Dimension: it has been closed in the whole process of (3.11), (3.20), (3.21).

3.5 Conceptual Explanation

1. Helical flow strength Q: geometrically defined spatial divergence rate, ensuring radial flux conservation.
2. Axial component v_z: automatically generated by continuity, no external "force" is required.
3. Space–mass coupling constant μ_g: binds geometric scale l_Ω with mass M, thereby fixing G.
4. Essence of axial infall: an axial compensation of spatial helical flow, not an independent gravitational action.

Summary of Part III

Through rigorous fluid-dynamical principles (continuity + Euler equations), we derive:

- Helical flow inevitably produces axial contraction.
- Infall acceleration naturally takes the inverse-square form $1/r^2$.
- Newton's law of gravitation is thus a necessary consequence of helical flow geometry and space–mass coupling, not an empirical postulate.

Explanation: the constant given by Bernoulli's "local two-point calibration" and "global Gauss bridge calibration" is only one geometric normalization difference; In this paper, the $\sigma = 4\pi$ in the "geometric calibration" specification is completely consistent with the Gaussian flux and spherical symmetry, so as to ensure a consistent closed loop with comparable dimensions and observations.

Part IV Observations and Numerical Calculations

4.1 Constants and Astronomical Parameters (values used in this section)

- Speed of light: $c = 2.99792458 \times 10^8$ m/s
- Gravitational constant: $G = 6.67430 \times 10^{-11}$ m^3/(kg·s^2)
- Earth: $GM_\oplus = 3.986004418 \times 10^{14}$ m^3/s^2, $meanradius\ R_\oplus = 6.371 \times 10^6$ m, total mass $M_\oplus = 5.97219 \times 10^{24}$ kg.
- Sun: $GM_\odot = 1.32712440018 \times 10^{20}$ m^3/s^2, $radius\ R_\odot = 6.9634 \times 10^8$ m, total mass $M_\odot = 1.9885 \times 10^{30}$ kg.
- Mercury's orbit: semi-major axis $a = 5.790905 \times 10^{10}$ m, eccentricity $e = 0.205630$, orbital period $T = 87.969$ d.

Note: In our geometric framework:

$$Q = 4\pi c \mathrm{l}_\Omega^2, \quad M = \mu_g \mathrm{l}_\Omega, \quad GM = \frac{c}{\mathrm{l}_\Omega} Q = 4\pi c^2 \mathrm{l}_\Omega$$

thus

$G = \frac{4\pi c^2}{\mu_g} \Rightarrow \mu_g = \frac{4\pi c^2}{G} \approx 1.69218 \times 10^{28}$ kg/m (constant, does not vary with celestial bodies)

4.2 Earth's Surface Gravitational Acceleration $g_\oplus$ (calculated directly by geometric calibration)

4.2.1 From Geometric Scaling to g

The geometric bridging relation gives $GM = 4\pi c^2 \mathrm{l}_\Omega$. At radius r, the axial infall acceleration is:

$$g(r) = -\frac{GM}{r^2}.$$

At Earth's surface $(r = R_\oplus)$:

$$g_\oplus = \frac{GM_\oplus}{R_\oplus^2} = \frac{4\pi c^2 \mathrm{l}_{\Omega,\oplus}}{R_\oplus^2} \quad (4.1)$$

4.2.2 Scaling Length and Numerical Result

$$\mathrm{l}_{\Omega,\oplus} = \frac{GM_{\oplus}}{4\pi c^2} = \frac{3.986004418 \times 10^{14}}{4\pi(2.99792458 \times 10^8)^2} \approx 3.5293 \times 10^{-4}\ \mathrm{m}$$

Substituting into Eq. (4.1):

$$g_{\oplus} = \frac{4\pi c^2\, \ell_{\Omega,\oplus}}{R_{\oplus}^2} = \frac{GM_{\oplus}}{R_{\oplus}^2} \approx \frac{3.986 \times 10^{14}}{(6.371 \times 10^6)^2} \approx 9.82\ m/\mathrm{s}^2$$

This matches the observed mean surface gravity (~9.8 m/s²). The value of μ_g is independent of celestial body, showing the non-rescaling property.

From $M = \mu_g \ell_{\Omega}$, the earth $\ell_{\Omega,\oplus}$ can be inversely solved, or the same result can be obtained with $M_{\oplus}$; The value of μ_g has nothing to do with celestial bodies (locked by G and c), reflecting "no rescaling".

4.3 Sun's Surface "Apparent Gravity" and Geometric Length

4.3.1 Solar Surface Acceleration

$$g_{\odot} = \frac{GM_{\odot}}{R_{\odot}^2} = \frac{1.32712440018 \times 10^{20}}{(6.9634 \times 10^8)^2} \approx 2.737 \times 10^2\ \mathrm{m/s}^2 \approx 274\ \mathrm{m/s}^2 \quad (4.1)$$

4.3.2 Solar Scaling Length and μ_g

$$\mathrm{l}_{\Omega,\odot} = \frac{GM_{\odot}}{4\pi c^2} \approx \frac{1.3271 \times 10^{20}}{4\pi(2.998 \times 10^8)^2} \approx 1.175 \times 10^2\ \mathrm{m} \quad (4.2)$$

Using $M_{\odot} = \mu_g \mathrm{l}_{\Omega,\odot}$ again verifies $\mu_g \approx 1.69218 \times 10^{28}$ kg/m.

Thus, different celestial bodies yield different l_{Ω}, but fixed μ_g → fixed G.

4.4 Mercury's Perihelion Precession

In our geometric–axial infall framework, at Newtonian leading order we recover $\Phi = -GM/r$. Including "curvature correction of helical geometry" yields a first-order effective potential perturbation equivalent to the standard scalar-curvature correction. The precession formula matches the 1PN (post-Newtonian) term of GR:

$$\Delta\varpi_{\text{per orbit}} = \frac{6\pi GM_{\odot}}{a(1-e^2)c^2} \quad (4.3)$$

Note: in our geometric model, this coefficient comes from the first-order geometric correction of "spin radial potential flow coupling of spiral flow"; Its form is consistent with the 1PN prediction of GR, which is convenient for direct comparison with observations.

4.4.1 Substitution of Mercury's Data

- $a = 5.790905 \times 10^{10}$ m, $\quad e = 0.205630$
- $GM_{\odot} = 1.32712440018 \times 10^{20}$ m^3/s^2

Per orbit precession (radians):

$$\Delta\varpi = \frac{6\pi(1.3271 \times 10^{20})}{(5.790905 \times 10^{10})(1 - 0.205630^2)(2.99792458 \times 10^8)^2}$$
$$\approx 5.017 \times 10^{-7} \text{ rad}$$

Convert to arcseconds:

$$\Delta\varpi \approx 0.1035 \text{ arcsec/orbit.}$$

Number of revolutions of mercury per century (Julian century=36525 days):

$$N = \frac{36525}{87.969} \approx 415.2 \ \ rail/century$$

so the total precession is:

$$\Delta\varpi_{\text{century}} \approx 0.1035 \times 415.2 \approx 43.0 \text{ arcsec/century} \qquad (4.4)$$

This exactly matches the observed value (~43″/century).

Key point: This result is not an independent assumption, but is based on Newton's $\Phi = -GM/r$, corresponding to our first-order correction of spiral geometry (velocity curvature coupling); Therefore, it can be regarded as the equivalent restatement and numerical verification of GR first-order effect in the context of "geometry fluid".

4.5 Black Hole Scales (Geometric Length and Schwarzschild Radius)

From the bridging relation $GM = 4\pi c^2 \mathrm{l}_\Omega$, the Schwarzschild radius is:

$$r_s = \frac{2GM}{c^2} = 8\pi \mathrm{l}_\Omega \qquad (4.5)$$

This directly links black hole radius with the geometric scaling length.

4.5.1 Solar-Mass Black Hole

$$r_{s,\odot} = \frac{2GM_\odot}{c^2} \approx \frac{2(1.3271 \times 10^{20})}{(2.998 \times 10^8)^2} \approx 2.953 \times 10^3 m \approx 3.0\,\mathrm{km}$$

Using $\mathrm{l}_{\Omega,\odot} \approx 117.5\,\mathrm{m}$ to verify (4.5):

$8\pi \ell_{\Omega,\odot} \approx 8\pi \times 117.5\, m \approx 2953\, m$， in perfect agreement.

4.5.2 Earth-Mass “Equivalent Schwarzschild Radius” (for scaling only)

$$r_{s,\oplus} = \frac{2GM_\oplus}{c^2} \approx 8.87 \times 10^{-3}\,\mathrm{m} \approx 8.9\,\mathrm{mm}$$

This is not a "real black hole", but a geometric length scale corresponding to the mass of the earth; The event horizon is formed only when the actual radius R is compressed to within r_s.

4.6 Summary (From Observation to Geometry)

1. Surface gravities:

$$g = \frac{GM}{R^2} = \frac{4\pi c^2 \mathrm{l}_\Omega}{R^2}$$

Numerical calculation results: $g_\oplus \approx 9.82\, m/\mathrm{s}^2$, $g_\odot \approx 274\, m/\mathrm{s}^2$, which are consistent with the observation.

2. Mercury perihelion precession:

Geometric first-order correction gives $\Delta\varpi = 6\pi GM/(a(1-e^2)c^2)$, the value $\approx$ 43"/century, which is consistent with the observation.

3. Black hole scales:

$$r_s = 2GM/c^2 = 8\pi\ell_\Omega$$

the black hole scale is linearly correlated with the geometric calibration length ℓ_Ω, showing the direct bridge of "geometric calibration → gravitational radius".

Part V Conclusions

1. Geometric scaling locks gravitational strength

From the helical flow source strength $Q = 4\pi c \mathrm{l}_\Omega^2$ and space–mass coupling $M = \mu_g \mathrm{l}_\Omega$, we derive:

$$G = \frac{4\pi c^2}{\mu_g}$$

showing that the gravitational constant G is no longer an empirical fitting parameter, but a geometric necessity.

2. Earth and Sun gravity calculations agree with observations

- Earth: with $\mathrm{l}_{\Omega,\oplus} \approx 3.5 \times 10^{-4}$ m, we obtain

$$g_\oplus = \frac{GM_\oplus}{R_\oplus^2} \approx 9.82\ \mathrm{m/s^2}$$

consistent with the measured value.

- Sun: with $l_{\Omega,\odot} \approx 1.2 \times 10^2$ m, we derive

$$g_\odot \approx 274\,\mathrm{m/s^2}$$

in perfect agreement with observations.

3. Geometric explanation of Mercury's perihelion precession

The first-order correction of helical geometry and axial infall gives

$$\Delta\varpi = \frac{6\pi G M_\odot}{a(1-e^2)c^2}$$

yielding $43''$/century, in exact agreement with observational data. This shows that precession is not an additional hypothesis but an intrinsic curvature effect of helical spatial flow.

4. Closed loop between black hole scale and geometric length

From the bridging relation $GM = 4\pi c^2 l_\Omega$, we derive

$$r_s = \frac{2GM}{c^2} = 8\pi l_\Omega$$

indicating that the Schwarzschild radius is directly proportional to the geometric scaling length. For example, a solar-mass black hole yields

$$r_s \approx 3.0\,\mathrm{km}$$

matching the standard result.

5. Core significance

- The inverse-square law of gravity $1/r^2$, planetary orbital dynamics, perihelion precession, and black hole radii all emerge from the same geometric–fluid

framework.

- The constant μ_g guarantees non-rescaling, fixing the relationship between G and c.
- This demonstrates that gravitation originates from the continuity of spatial helical flow and space–mass coupling. Axial infall is thus a geometric property of spacetime, not an externally imposed mechanical assumption.

Final statement:

This paper proves that gravitation is not an independent postulate, but the inevitable axial infall generated by outward helical expansion of space at the speed of light. From geometric scaling and space–mass coupling, we obtain $G = 4\pi c^2/\mu_g$, thereby unifying the explanations of Earth's and Sun's surface gravity, Mercury's perihelion precession, and black hole radii. The results show that both the inverse-square law of gravity $1/r^2$ and extreme astrophysical scales originate from the closed-loop structure of geometric helical flow, revealing the true geometric origin of gravitation.

Appendix

A.1 Symbol Table (Chinese–English Correspondence)

Symbol	Definition	Dimension	English Term
c	Speed of light	m/s	speed of light
l_Ω	Rotational scaling length	m	rotational scaling length
Q	Helical flow strength (spatial divergence source)	m^3/shelical	divergence strength
μ_g	Space–mass coupling constant	kg/m	space–mass coupling constant
M	Coupled mass	kg	coupled mass
G	Gravitational constant	$m^3/(kg\cdot s^2)$	gravitational constant

g	Radial/axial infall acceleration	m/s²infall acceleration	
Φ	Geometric potential function	m²/s²	geometric potential
r_s	Schwarzschild radius m	Schwarzschild radius	

A.2 Formula Summary (Mother Formulas and Derivation Chain)

1. Helical flow strength

$$Q = 4\pi c \mathrm{l}_\Omega^2, \quad [Q] = m^3/s \quad (A1)$$

2. Space–mass coupling

$$M = \mu_g \mathrm{l}_\Omega, \quad [\mu_g] = kg/m \quad (A2)$$

3. Geometric expression of gravitational constant

$$G = \frac{4\pi c^2}{\mu_g}, \quad [G] = m^3/(kg \cdot s^2) \quad (A3)$$

4. Newtonian gravitational form (infall acceleration)

$$g = -\frac{GM}{r^2}, \quad [g] = m/s^2 \quad (A4)$$

5. Mercury's perihelion precession (geometric first-order correction)

$$\Delta\varpi = \frac{6\pi GM}{a(1-e^2)c^2}, \quad [\Delta\varpi] = \text{rad/orbit} \quad (A5)$$

6. Relation between black hole radius and geometric scaling

$$r_s = \frac{2GM}{c^2} = 8\pi \mathrm{l}_\Omega \quad (A6)$$

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